

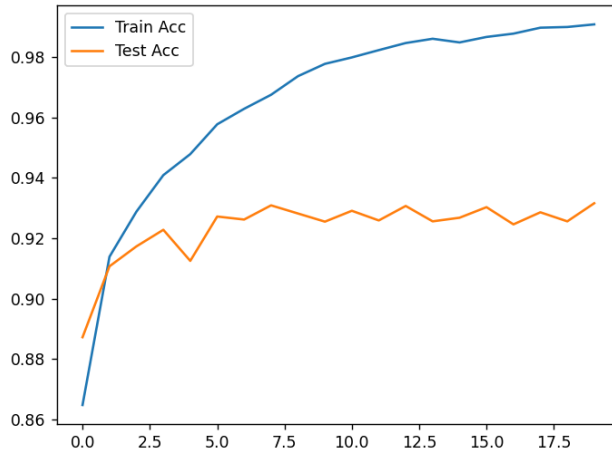
인공지능개론 미니프로젝트

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01. PyTorch에서 제공하는 CNN을 이용한 패션 MNIST 데이터셋을 분류하는 모델 구현

Figure 1

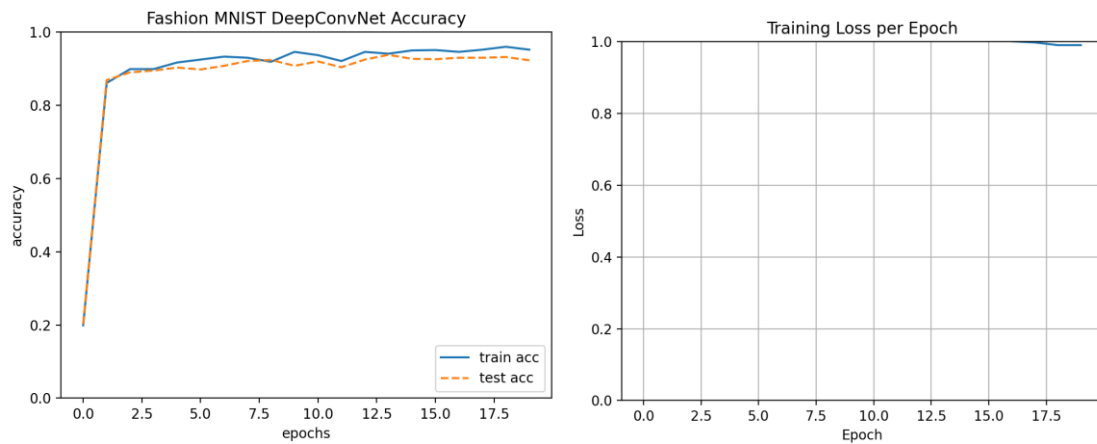


Home Left Right Zoom In Zoom Out Print

===== RESTART: C:/Users/82106/Documents/가은 대학/3-1/인공지능개론/fashion_cnn.py =====

Epoch 1: Train Acc=0.8647, Test Acc=0.8872
Epoch 2: Train Acc=0.9139, Test Acc=0.9107
Epoch 3: Train Acc=0.9288, Test Acc=0.9173
Epoch 4: Train Acc=0.9409, Test Acc=0.9228
Epoch 5: Train Acc=0.9479, Test Acc=0.9125
Epoch 6: Train Acc=0.9578, Test Acc=0.9272
Epoch 7: Train Acc=0.9629, Test Acc=0.9262
Epoch 8: Train Acc=0.9676, Test Acc=0.9309
Epoch 9: Train Acc=0.9737, Test Acc=0.9282
Epoch 10: Train Acc=0.9778, Test Acc=0.9255
Epoch 11: Train Acc=0.9799, Test Acc=0.9291
Epoch 12: Train Acc=0.9823, Test Acc=0.9259
Epoch 13: Train Acc=0.9847, Test Acc=0.9307
Epoch 14: Train Acc=0.9861, Test Acc=0.9256
Epoch 15: Train Acc=0.9849, Test Acc=0.9268
Epoch 16: Train Acc=0.9867, Test Acc=0.9303
Epoch 17: Train Acc=0.9878, Test Acc=0.9246
Epoch 18: Train Acc=0.9898, Test Acc=0.9286
Epoch 19: Train Acc=0.9900, Test Acc=0.9256
Epoch 20: Train Acc=0.9909, Test Acc=0.9316

02. CNN 소스코드를 이용하여 패션 MNIST 데이터셋을 분류하는 모델을 구현



```

train loss:1.0520454058200541
train loss:1.1289260630456985
train loss:1.0077862728878197
train loss:0.8773565196681988
train loss:1.006958590992369
train loss:1.0059600127132986
train loss:0.9737312831124143
train loss:1.0190904488683248
train loss:1.0737532319928944
train loss:1.1156775977416968
train loss:0.8569805768664781
train loss:1.0986923458691467
train loss:0.9889501965182466
train loss:1.0986955376771497
train loss:0.9518795596671892
train loss:0.8696837334258444
train loss:0.8207494116079974
train loss:0.8956548985142696
train loss:0.8510189221445619
===== Final Test Accuracy =====
test acc:0.9193
--- Final Results ---
Train Acc: 95.2 % Test Acc: 92.3 %

```